

Parallel Traffic Density Simulation with Multi-Lane and Weather Effects using Matrix-Based Computation

In real cities, traffic conditions change continuously due to:

- Number of vehicles on roads
- Multiple lanes on each road
- External conditions such as rain or accidents



Real World → Computational Model

Real World	Computational Model
City road network	2D grid (matrix)
Road segment	One cell in the matrix
Cars on the road	Traffic density value
Multiple lanes	Extra matrix dimension
Weather / accidents	Modifier values
Time passing	Iterative updates

Traffic Update Logic

1. Read current traffic density
2. Read neighboring road densities
3. Compute base traffic update
4. Apply weather modifier
5. Store updated traffic density



Serial Execution → One Traffic Engineer Doing Everything

A small town has:

- 200 road segments (rows)
- 200 intersections (columns)
- 3 lanes per road

The town hires only one traffic engineer to monitor traffic conditions.

Road Map

|— Segment (0,0) Lane 1
|— Segment (0,0) Lane 2
|— Segment (0,0) Lane 3
|— Segment (0,1) Lane 1
|— Segment (0,1) Lane 2
|— Segment (0,1) Lane 3

...

|— Segment (199,199) Lane 3

Everything is processed in order, one after another.



MPI → Multiple Cities / Regions

Imagine a national traffic authority.

- The country is divided into regions or cities
- Each region has its own traffic control center
- Each center is responsible only for its region's roads

Example:

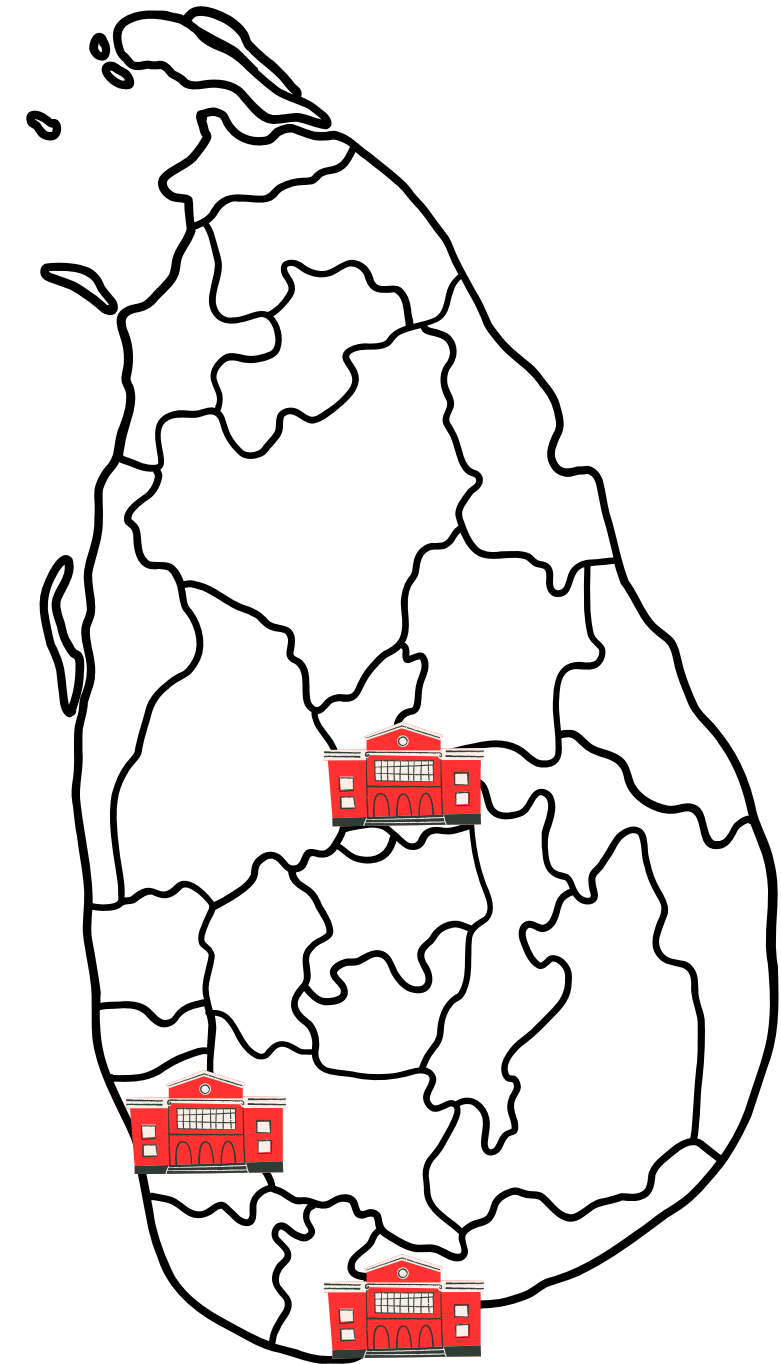
Colombo Control Center → Colombo roads

Kandy Control Center → Kandy roads

Galle Control Center → Galle roads

This is MPI:

- Each control center = one MPI process
- Each process has its own memory



OpenMP → Engineers inside one Control Center

Inside each traffic control center:

- There are multiple traffic engineers
- All engineers work on the same city map
- Each engineer updates different road segments or lanes

Example:

Colombo Control Center:

Engineer 1 → Main roads

Engineer 2 → Side roads

Engineer 3 → Highway lanes

Engineers:

- Share the same information
- Do not send messages to each other
- Just work in parallel



Hybrid Model = Both Levels Together

Put both together:

Country

- |— Colombo Control Center (MPI Process)
 - | |— Engineer 1 (OpenMP Thread)
 - | |— Engineer 2 (OpenMP Thread)
- |— Kandy Control Center (MPI Process)
 - | |— Engineer 1 (OpenMP Thread)
 - | |— Engineer 2 (OpenMP Thread)

- MPI divides work across regions
- OpenMP divides work within each region

